👩‍🏫 Staff Guide · Ag Science 1 · Dairy (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. Same topics and order as Mr. Torretta's class; more reading and writing than Level 2.

1. How the pieces line up

Class lesson	Level 3 piece
Dairy Production (Blank Out notes)	Reading parts 1-3; notes section A
Darigold video	Reading part 4; notes section B
Dairy Products notes; Cheese presentation; Greek Yogurt; Steps of Ice Cream	Reading part 5; notes section C
Unit review	Quick check + unit game

2. Supports built in

- Reading and notes each have a web copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of the notes; word problems have a worked example first
- Less filled in than Level 2: short answers, tables, a drawing
- No slide decks: students follow Mr. Torretta's slides with the Level 3 notes

3. Files

- Reading (+ 🔊 web copy)
- Guided notes + key (+ 🔊 web copy)
- Quick check + key
- Unit game: 10 questions

4. What to grade

- Guided notes stand in for the matching class notes or worksheet (same points when complete)
- Quick check: score it, then scale to the class quiz points
- Reading check questions: completion points
- Modified grading only as the IEP allows. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Class content: Mr. Torretta.

🐄 Dairy: Guided Notes

Ag Science 1 · Unit 6 · Level 3 (closer to the class work) · goes with Dairy Production + Dairy Products notes, Darigold video, Cheese presentation, Greek Yogurt, Steps of Ice Cream

Name: _____

Use the class slides and your reading. Key words:

Holstein | *Jersey* | *colostrum* | *oxytocin* | *milking parlor* | *pasteurize* | *homogenize* | *casein* | *overrun*

A. Dairy production (Blank Out notes)

1. Milk is Washington's # _____ farm product.
2. The black-and-white breed with the most milk is the _____.
3. Milk let-down is caused by the hormone _____.
4. **Percent practice**

> Example: milk is 87% water, so 10 lb of milk has 10 × 0.87 = 8.7 lb of water

a) Water in 100 lb of milk = _____

b) A herd of 100 cows gives 2,300,000 lb a year. Pounds per cow = _____

B. Farm to plant

5. Fill in the table.

Step	What happens
Truck check	
Pasteurize	
Homogenize	

6. Why must pasteurized milk stay in the fridge?

_____

_____

C. Dairy products (Cheese, Greek yogurt, Ice cream)

7. Cheese forms when _____ clumps together.
8. Ice cream must have at least _____ % milk fat.
9. How is Greek yogurt different from regular yogurt?

_____

_____

🐄 Dairy: Reading

Ag Science 1 · Unit 6 · Level 3 (closer to the class work) · goes with Dairy Production + Dairy Products notes, Darigold video, Cheese presentation, Greek Yogurt, Steps of Ice Cream

Name: _____

From Cow to Carton

Milk and dairy products are Washington's number two farm product, after apples; milk was worth about $1.2 billion in 2017. The U.S. has about 9.3 million dairy cows. In 50 years the number of dairy cows dropped more than 50%, but milk per cow rose more than 130%.

The Holstein, black and white, makes up about 80% of registered U.S. dairy cattle and gives the most milk. The small brown Jersey gives milk high in fat. Other breeds are Guernsey, Brown Swiss, Ayrshire, Red and White and Milking Shorthorn. A calf's first milk, colostrum, has antibodies that protect it, so calves need it soon after birth.

Milk is about 87% water, plus fat, protein (casein), lactose (milk sugar) and minerals. It is a key source of protein, calcium and riboflavin. When a cow's teat is touched, the brain releases oxytocin, which lets milk down. Pain, loud noise or stress can block let-down, so calm cows give more milk. Cows are milked by machine in a milking parlor.

The milk truck driver checks that milk is under 40°F, checks the smell and takes a sample to test for bacteria, fat and antibiotics. At the plant milk is pasteurized, heated to kill harmful bacteria (most often 161°F for 15 seconds), and homogenized, which breaks up fat so cream does not rise. Vitamin D is added. Pasteurized milk still must stay below 40°F.

Milk becomes many products. Cheese forms when casein clumps (coagulates). Yogurt contains probiotics; Greek yogurt is strained to be thicker. Butter is made from cream. Ice cream must have at least 10% milk fat, and the air whipped in is called overrun.

Check your reading

1. Why are there fewer cows but more milk than 50 years ago?

2. What is the difference between pasteurizing and homogenizing?

3. Why do calves need colostrum?

🐄 Dairy: Quick Check

Ag Science 1 · Unit 6 · Level 3 (closer to the class work) · goes with Dairy Production + Dairy Products notes, Darigold video, Cheese presentation, Greek Yogurt, Steps of Ice Cream

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. About 80% of registered dairy cows are…

A. Holsteins | B. Jerseys | C. Guernseys

2. Pasteurizing milk…

A. Adds fat | B. Makes cheese | C. Kills harmful bacteria with heat

3. Milk should be kept below…

A. 70°F | B. 100°F | C. 40°F

4. Colostrum is…

A. A cheese | B. The first milk after birth | C. A cow breed

5. Explain why calm cows give more milk.

6. Name two dairy products and how each is made.

🐄 Dairy: Guided Notes · ANSWER KEY

Ag Science 1 · Unit 6 · Level 3 (closer to the class work) · goes with Dairy Production + Dairy Products notes, Darigold video, Cheese presentation, Greek Yogurt, Steps of Ice Cream

Name: _____

Use the class slides and your reading. Key words:

Holstein | *Jersey* | *colostrum* | *oxytocin* | *milking parlor* | *pasteurize* | *homogenize* | *casein* | *overrun*

A. Dairy production (Blank Out notes)

1. Milk is Washington's # **2** farm product.
2. The black-and-white breed with the most milk is the **Holstein**.
3. Milk let-down is caused by the hormone **oxytocin**.
4. **Percent practice**

> Example: milk is 87% water, so 10 lb of milk has 10 × 0.87 = 8.7 lb of water

a) Water in 100 lb of milk = **87 lb**

b) A herd of 100 cows gives 2,300,000 lb a year. Pounds per cow = **23,000 lb**

B. Farm to plant

5. Fill in the table.

Step	What happens
Truck check	**under 40°F, smell, sample tested**
Pasteurize	**heat (161°F, 15 s) kills harmful bacteria**
Homogenize	**breaks up fat so cream doesn't rise**

6. Why must pasteurized milk stay in the fridge?
Pasteurizing does not sterilize; some bacteria remain and grow fast when warm.

C. Dairy products (Cheese, Greek yogurt, Ice cream)

7. Cheese forms when **casein** clumps together.
8. Ice cream must have at least **10** % milk fat.
9. How is Greek yogurt different from regular yogurt?
It is strained to remove liquid, so it is thicker (and higher in protein).

🔑 *Key written by our team from Mr. Torretta's materials, please check.*

🐄 Dairy: Reading · ANSWER KEY

Ag Science 1 · Unit 6 · Level 3 (closer to the class work) · goes with Dairy Production + Dairy Products notes, Darigold video, Cheese presentation, Greek Yogurt, Steps of Ice Cream

Name: _____

From Cow to Carton

Milk and dairy products are Washington's number two farm product, after apples; milk was worth about $1.2 billion in 2017. The U.S. has about 9.3 million dairy cows. In 50 years the number of dairy cows dropped more than 50%, but milk per cow rose more than 130%.

The Holstein, black and white, makes up about 80% of registered U.S. dairy cattle and gives the most milk. The small brown Jersey gives milk high in fat. Other breeds are Guernsey, Brown Swiss, Ayrshire, Red and White and Milking Shorthorn. A calf's first milk, colostrum, has antibodies that protect it, so calves need it soon after birth.

Milk is about 87% water, plus fat, protein (casein), lactose (milk sugar) and minerals. It is a key source of protein, calcium and riboflavin. When a cow's teat is touched, the brain releases oxytocin, which lets milk down. Pain, loud noise or stress can block let-down, so calm cows give more milk. Cows are milked by machine in a milking parlor.

The milk truck driver checks that milk is under 40°F, checks the smell and takes a sample to test for bacteria, fat and antibiotics. At the plant milk is pasteurized, heated to kill harmful bacteria (most often 161°F for 15 seconds), and homogenized, which breaks up fat so cream does not rise. Vitamin D is added. Pasteurized milk still must stay below 40°F.

Milk becomes many products. Cheese forms when casein clumps (coagulates). Yogurt contains probiotics; Greek yogurt is strained to be thicker. Butter is made from cream. Ice cream must have at least 10% milk fat, and the air whipped in is called overrun.

Check your reading

1. Why are there fewer cows but more milk than 50 years ago?
Each cow gives much more milk (better breeding, feed and care).

2. What is the difference between pasteurizing and homogenizing?
Pasteurizing heats milk to kill harmful bacteria; homogenizing breaks up fat.

3. Why do calves need colostrum?
It has antibodies that protect the newborn calf.

🐄 Dairy: Quick Check · ANSWER KEY

Ag Science 1 · Unit 6 · Level 3 (closer to the class work) · goes with Dairy Production + Dairy Products notes, Darigold video, Cheese presentation, Greek Yogurt, Steps of Ice Cream

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. About 80% of registered dairy cows are…

A. Holsteins | B. Jerseys | C. Guernseys

2. Pasteurizing milk…

A. Adds fat | B. Makes cheese | **C. Kills harmful bacteria with heat**

3. Milk should be kept below…

A. 70°F | B. 100°F | **C. 40°F**

4. Colostrum is…

A. A cheese | **B. The first milk after birth** | C. A cow breed

5. Explain why calm cows give more milk.

Stress, pain or noise blocks oxytocin, which stops milk let-down.

6. Name two dairy products and how each is made.

Any two: cheese (casein clumps), butter (from cream), yogurt (cultured with probiotics), ice cream (≥10% fat, frozen with air).

🔑 *Key written by our team from Mr. Torretta's quizzes and notes, please check.*